Name: Date: Period:

Coulomb's Law

1. A single electron is placed 3.0 nm to the right of a 0.4 nC charge. Calculate the *magnitude* and *direction* of the force acting on the electron.

2. Three point charges q_1, q_2, and q_3 lie along the x-axis at $x = -0.4$ m, 0 m, and 1.0 m, respectively. Calculate the magnitude and direction of electrostatic force on charge q_3 when

$$q_1 = +1.3 \times 10^{-5}\,\mathrm{C}\,,$$
$$q_2 = -6.0 \times 10^{-5}\,\mathrm{C}\,, \text{ and}$$
$$q_3 = -5.7 \times 10^{-5}\,\mathrm{C}\,.$$